ECE
Probabilistic Systems Analysis

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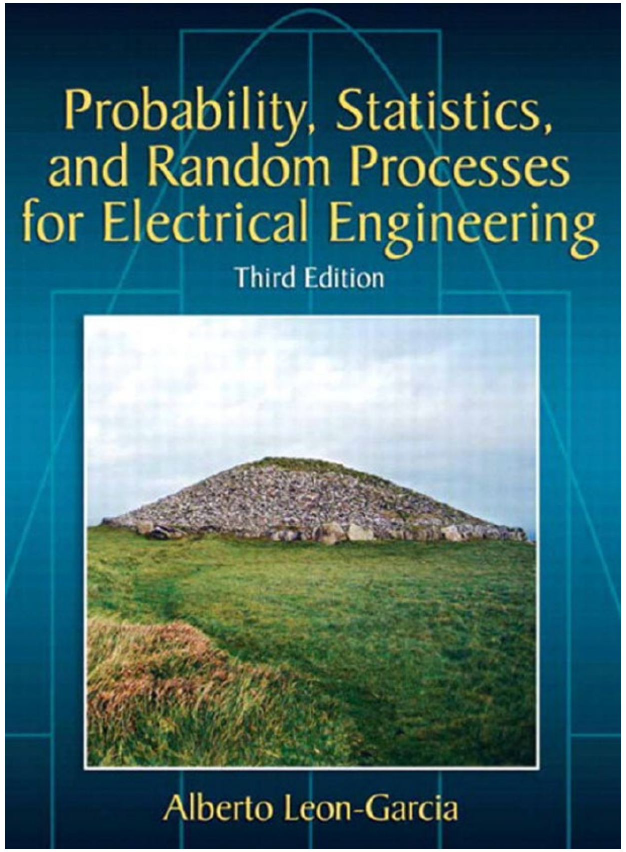
Introduction



Lectures

1. Probability Models in Electrical and Computer Engineering
2. Basic Concepts of Probability Theory
3. Discrete Random Variables
4. One Random Variable
5. Pairs of Random Variables
6. Vector Random Variables
7. Sums of Random Variables and Long-Term Averages
8. Statistics
9. Random Processes

Prerequisite

1. Calculus I
2. Matlab Programming

Textbooks

Alberto Leon-Garcia "Probability, Statistics, and Random Processes For Electrical Engineering", 3rd Edition, 2008.

Grading

Mid-term Exam	30
Quizzes	8
MATLAB Assignments	8
Attendance	4
Final Exam	50

Lecture Outline

- Probability as a mathematical framework for:
  - reasoning about uncertainty
  - developing approaches to inference problems
- Probabilistic models
  - sample space
  - probability law
- Axioms of probability
- Simple examples

Mathematical Model

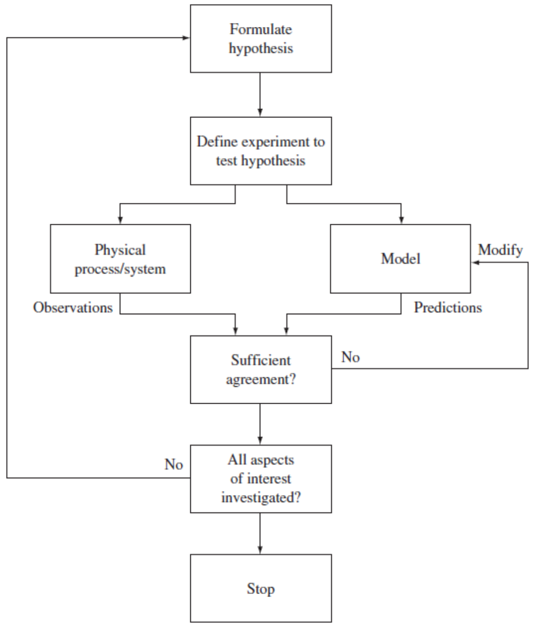
A **model** is an approximate representation of a physical situation.

Deterministic *Model*

The solution of a set of mathematical equations specifies the exact outcome of the experiment.

Probability Model

Probability models are one of the tools that enable the designer to make sense out of the chaos and to successfully build systems that are efficient, reliable, and cost effective.

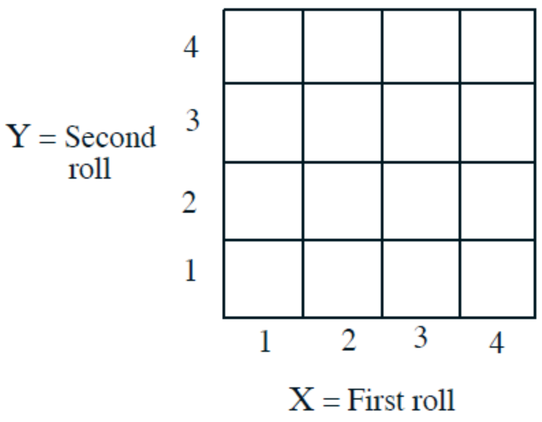


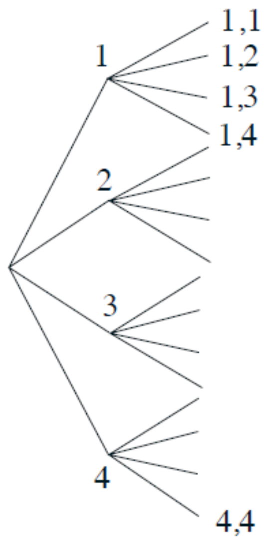
Sample Space

- “List” (set) of possible outcomes
- List must be:
  - Mutually exclusive
  - Collectively exhaustive
- Art: to be at the “right” granularity

Sample space: Discrete example

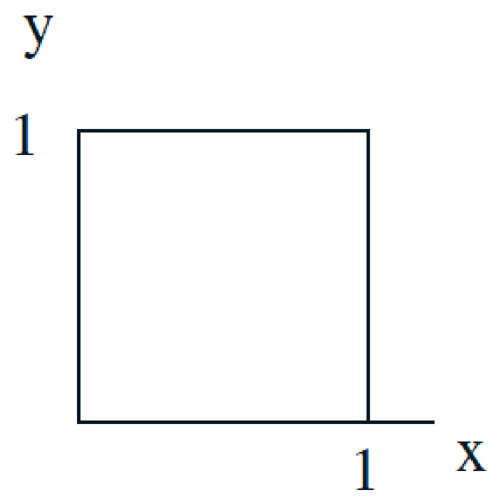
- Two rolls of a tetrahedral die
  - Sample space vs. sequential description





Sample space: Continuous example

$$\Omega = \{(x, y) \mid \ 0 \le x, y \le 1\}$$



Probability axioms

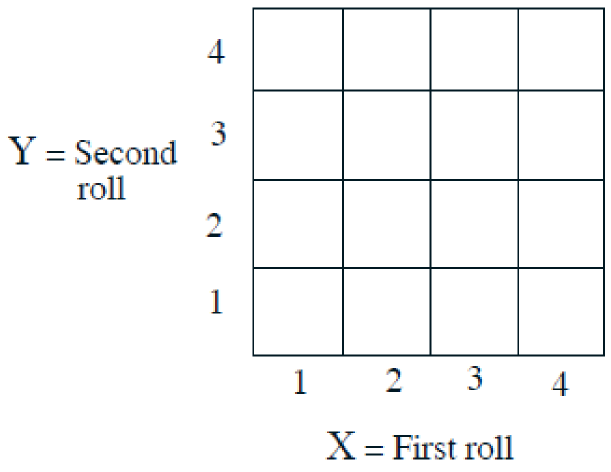
- **Event:** a subset of the sample space
- Probability is assigned to events

Axioms:

1. **Nonnegativity:** $\mathbf{P}(A) \geq 0$
2. **Normalization:** $\mathbf{P}(\Omega) = 1$
3. **Additivity:** If $A \cap B = \emptyset$, then $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$

- $$\begin{aligned} \mathbf{P}(\{s_1, s_2, \ldots, s_k\}) &= \mathbf{P}(\{s_1\}) + \cdots + \mathbf{P}(\{s_k\}) \\ &= \mathbf{P}(s_1) + \cdots + \mathbf{P}(s_k) \end{aligned}$$

Example with finite sample space



- Let every possible outcome have probability 1/16
  - $\mathrm{P}((X, Y)$ is (1,1) or (1,2)$) =$
  - $\mathrm{P}(\{X = 1\}) =$
  - $\mathrm{P}(X + Y$ is odd$) =$
  - $\mathrm{P}(\min(X, Y) = 2) =$

Discrete uniform law

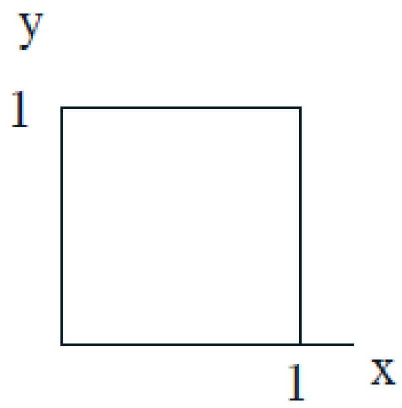
- Let all outcomes be equally likely
- Then,

$$P(A) = \frac{\text{number of elements of } A}{\text{total number of sample points}}$$

- Computing probabilities $\equiv$ counting
- Defines fair coins, fair dice, well-shuffled card decks

Continuous uniform law

- Two "random" numbers in $[0, 1]$.



- **Uniform** law: Probability = Area

  – $\mathrm{P}(X + Y \leq 1/2) = ?$

  – $\mathrm{P}((X, Y) = (0.5, 0.3))$